Project Initialization and Planning Phase

Date	25-Jun-2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

OptiCrop is an AI-powered crop recommendation system that uses machine learning to suggest the most suitable crops based on soil nutrients and environmental conditions. It helps farmers make informed decisions, improve crop productivity, optimize resource utilization, and reduce the risk of crop failure through a simple web-based application.

Project Overview	
Objective	The primary objective of OptiCrop is to assist farmers in selecting the most suitable crop by analyzing soil nutrients (Nitrogen, Phosphorus, Potassium) and environmental factors such as temperature, humidity, pH, and rainfall using machine learning techniques.
Scope	The project focuses on collecting agricultural data, preprocessing the dataset, training a machine learning model, and developing a Flask-based web application that provides real-time crop recommendations to farmers and agricultural stakeholders.
Problem Statement	
Description	Farmers often struggle to choose the right crop because of changing soil conditions, varying weather patterns, and limited access to expert agricultural guidance. Traditional farming decisions may result in unsuitable crop selection, low productivity, inefficient resource utilization, and financial losses.

Impact	Providing intelligent crop recommendations will improve agricultural productivity, optimize the use of water and fertilizers, reduce farming risks, increase profitability, and support sustainable farming practices.
Proposed Solution	
Approach	Develop a machine learning-based crop recommendation system that analyzes soil nutrients and environmental conditions to predict the most suitable crop for cultivation. The trained model will be integrated into a Flask web application for real-time recommendations.
Key Features	• AI-powered crop recommendation using Machine Learning. • Real-time prediction based on soil and weather parameters. • Crop suitability assessment for informed decision-making. • Interactive and user-friendly web interface. • Future support for fertilizer recommendation, disease detection, and yield prediction. - Real-time decision-making for quicker loan approvals. -Continuous learning to adapt to evolving financial landscapes.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/i7 Processor or equivalent
Memory	RAM specifications	Minimum **8 GB RAM**
Storage	Disk space for data, models, and logs	**256 GB SSD** or higher
Software		
Frameworks	Python frameworks	Flask (Python)

Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Joblib
Development Environment	IDE	Visual Studio Code, Jupyter Notebook
Data		
Data	Source, size, format	Kaggle Crop Recommendation Dataset, approximately **2,200 records**, CSV format